T4 GPU vs 8-Core CPU + 24GB RAM

1. Modal Pricing (2026)

Resource	Per Second	Per Hour	Notes
T4 GPU	$0.000164	$0.59	16GB VRAM
CPU (1 core)	$0.000039	$0.14	2 vCPU = 1 core
Memory (1GB)	$0.0000022	$0.008	GiB/sec

2. Total Hourly Cost

Config	Components	Total/hr	Monthly (24/7)
T4 GPU	GPU only	$0.59	$424.80
8 CPU + 24GB	CPU + Memory	$1.33	$957.60

8 CPU + 24GB costs 2.3x MORE per hour than T4 GPU.

3. Inference Time Comparison

Config	Model Load	Per Call	Total
T4 GPU	5-8 sec	4-8 sec	8-16 sec
8 CPU + 24GB	15-25 sec	2-4 min	2.5-4.5 min

T4 is 17x FASTER per inference call.

4. Per-Call Cost (THE KEY METRIC)

Cost per inference = (Hourly cost / 3600) x Time per call

Config	Cost/hr	Time/Call	Cost/Call	Monthly (100/day)
T4 GPU	$0.59	8 sec	$0.0013	$3.90
8 CPU + 24GB	$1.33	180 sec	$0.0665	$19.95

T4 is 51x CHEAPER per inference call.

5. Why CPU + RAM Is Slower for LLMs

LLMs rely on matrix multiplication which GPUs handle in parallel across thousands of cores. CPUs process these operations sequentially, making them 15-25x slower for transformer inference.

Operation	T4 GPU (2560 CUDA cores)	CPU (8 cores)
Matrix multiply	Parallel (ms)	Sequential (sec)
Memory bandwidth	320 GB/s	50 GB/s
Tensor cores	320 (FP16)	None

6. Summary

T4 GPU wins on BOTH cost and speed.

Metric	T4 GPU	8 CPU + 24GB	Winner
Hourly cost	$0.59	$1.33	T4 (2.3x cheaper)
Inference time	8 sec	180 sec	T4 (22x faster)
Cost per call	$0.0013	$0.067	T4 (51x cheaper)
Monthly (100/day)	$3.90	$19.95	T4 (5x cheaper)

Conclusion: There is NO scenario where CPU + RAM is better than T4 for LLM inference.